

Introduction to Git

Module 1: Introduction to Git

- What is Git?
 - Free and open-source distributed version control system.
 - Tracks changes in files.
 - Used for collaboration in software development.
 - Goals: speed, data integrity, distributed workflows.
 - Git vs Other Version Control Systems (brief comparison).
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Module 2: Installing and Configuring Git

- **Download Git:** git-scm.com/downloads
 - **Verify installation:**

```
git --version
```
 - **Configure Git (user details):**

```
git config --global user.name "John Kumar"
git config --global user.email "john@gmail.com"
git config --list
```
 - **Editor basics** (if Git opens Vim):
 - Exit: ESC → :wq
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Module 3: Creating a Repository

- **Create project folder & initialize Git:**

```
mkdir LetsLearnGit
cd LetsLearnGit
git init
```
 - **Check repository status:**

```
git status
```
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Module 4: Working with Files (Staging & Committing)

- **Add files to staging area:**

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```
git add <filename>
git add .      # add all files
```

- **Commit changes:**

```
git commit -m "Initial commit"
```

- **View commit history:**

```
git log
```

- **Key concept:** HEAD → points to the current branch.
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Module 5: Branching and Navigation

- **Create a branch:**

```
git branch dev
```

- **List all branches:**

```
git branch
```

- **Switch branch:**

```
git checkout dev
```

- **Shortcut (create & switch):**

```
git checkout -b mukesh/multiply
```

- **Checkout commit by hash:**

```
git checkout <commit-hash>
git checkout master    # return to latest
```

Module 6: Merging Branches

- **Merge changes:**

```
git merge <branch-name>
```

- **Example workflow:**

1. Create branch mukesh/multiply
2. Make changes (e.g., Multiply.java)
3. Switch back to dev

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4. Merge mukesh/multiply into dev
 5. Merge dev into master
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Module 7: Ignoring Files

- **Create** `.gitignore` file to exclude unwanted files.

Example:

```
*.class  
/target/  
secret.txt
```

- Verify with `git status`.
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Module 8: Practical Demo Workflow

1. Create repository `LetsLearnGit`.
 2. Add `Sum.java`, commit it.
 3. Modify `Sum.java`, create new file `Diff.java`.
 4. Stage only `Diff.java`, commit → observe `Sum.java` untracked.
 5. Stage both files, commit again.
 6. Explore `git log` with different commits.
 7. Checkout previous commits using hash.
 8. Create & switch branches → merge changes.
 9. Use `.gitignore` for excluding files.
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